

A Higher Order Rao-Blackwellized Particle Filter for Monocular vSLAM

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Abstract— This paper generalizes the traditional formulation of Rao-Blackwellized particle filter (RBPF) by incorporating a higher order state variable and a modified undelayed initialization scheme to solve the 3D monocular visual SLAM problem (vSLAM). In the proposed approach, velocity has been included in the state variables so that filtering progresses based on sampling from velocity distribution, not the displacement. The new sampling posterior has been obtained with respect to observations, control inputs and the robot path. The proper importance weight for resampling has been derived in this paper. To solve the bearing-only problem, the proposed approach features a modified initialization scheme that uses an inverse depth of the landmarks. The proposed higher order RBPF approach has been compared to the traditional (lower order) RBPF approach for proof of concept through a tangible simulation routine. The results of the numerical simulation indicate the superiority of the higher order RBPF in certain conditions e.g., high parallax angles.

I. INTRODUCTION

THE core idea of SLAM is to build a map incrementally and simultaneously locate the robot in the map by measuring the robot control input and measurements. Since the control and sensing signals suffer from noise, the SLAM problem requires a probabilistic representation. Smith et al. [1] established a probabilistic approach for describing robot motion, landmark locations and the corresponding uncertain geometry. In this approach, the robot *observation model* and *motion model* are presented as the zero-order and first-order Markov processes, respectively. In general, these processes can be nonlinear and non Gaussian. Using this probabilistic approach, the SLAM problem reduces to a filtering problem in which the control input and robot observations are known, and the instantaneous robot pose and landmark locations are desired unknown states. Seeking a solution to SLAM problem, Csorba and Durrant-Whyte [2] proposed the use of the Extended Kalman Filter (EKF) to solve this nonlinear filtering problem. Their work was further extended in [3] where convergence and performance of the EKF-solution was discussed, and a successful experiment was reported.

In the EKF algorithm, the covariance matrix maintains

all N landmarks, and hence requires an $O(N^2)$ space. This covariance matrix has to be updated with each measurement at a computational cost proportional to $O(N^2)$. This computational cost allows real-time implementation only for sparse mapping not for large environments or dense maps. Various efforts have been made to reduce computational load of the EKF SLAM. Liu and Thrun proposed a new solution the SLAM problem using the Extended Information Filter (EIF) [4]. In this solution, the information matrix can be sparsified that can save the memory and reduce computation load. Another solution for the SLAM problem, FastSLAM, based on the Rao-Blackwellized particle filtering (RBPF) was presented by [5]. FastSLAM is based on factorization of the robot path from landmark locations in which observation of each landmark is independent of any other landmark position [6]. Using particle filters instead of the Kalman filter results in less computational complexity and more robustness in data association [7]. Also, unlike the EKF, there is no need for the linearization of the motion model. Later, a modified version of FastSLAM, FastSLAM 2.0, with enhanced convergence and performance was proposed by Motemerlo et al. [8].

Contemporary research in this field has focused on inexpensive and computationally viable solutions for vSLAM that rely on the images of a monocular vision system i.e., a single camera, as the only source of information for generating the map [9, 10]. The use of a single camera, a bearing-only device, as the only source of information for the SLAM systems causes the problem of the partial observability [11]. In bearing-only devices such as a camera, sensors are not able to provide any information on the depth of the landmarks. It is noted that the standard SLAM solutions are based on a fully observable system.

In the bearing-only systems, as it is discussed in [12], the estimation system performance depends strongly on the system order. Higher order systems are able to track the more complicated maneuvers while they can increase the estimation error in the simpler ones. In the EKF based SLAM framework, the motion model order can be changed easily and additional states can be added to the state vector and covariance matrix. On the other hand, in the bearing-only systems using Rao-Blackwellized SLAM framework, higher order states have no effect in the observation. Therefore, FastSLAM 2.0 being superior in performance and convergence in comparison to the FastSLAM is not applicable to the bearing-only systems with higher order states, i.e. velocity or acceleration.

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In this paper, a velocity based RBPF approach for the SLAM of the bearing-only systems is proposed. In the proposed approach, first the robot pose is updated by sampling velocity from the proposal distribution. The proposed approach, similar to FastSLAM 2.0, utilizes information of both control input and observation to obtain the proposal distribution. This issue makes the proposed approach less wasteful in comparison with the time that it relies only on the control input. Considering updated robot pose and the last observation, the landmark locations are updated using the EKF. In the next stage, importance weight of each particle is calculated. In the last step the set of particles is resampled with respect to each particle weight. Also, to solve the problem of the partial observability, each feature is initialized using a modified scheme similar to the algorithm discussed in [13].

In the next section, details of the proposed approach including sampling and importance weight distribution are discussed. In Section III, the motion and observation model of a bearing-only system, i.e. a camera, is discussed. To evaluate the proposed algorithm performance, numerical simulations have been carried out which are presented in the Section IV. In the last section conclusion of the paper is presented.

II. VELOCITY BASED RBPF APPROACH

In the SLAM problem, the only available information for the observer is the measurements, $z_{1:t}$ and the control inputs $u_{1:t}$, e.g. acceleration, as they become available. The index $1:t$ refers to the time period between the first iteration and the iteration at time t . In the vision-based SLAM, it can be assumed that the data association problem [14] is solved using image processing methods such as the SIFT algorithm [15, 16]. The bearing-only system is modeled using a constant velocity model which means that both position and velocity are included in the state vector. In this way, the accelerations are characterized using a Gaussian distribution. The approximated acceleration is assumed as the control input. The velocity dynamics can be presented as the first order Markov chain:

$$p(x_t | u_t, x_{t-1}) \sim \mathcal{N}(x_t; f(x_{t-1}, u_t), R_t) \quad (1)$$

where x_t is the velocity in time t . The symbol $\mathcal{N}$ stands denotes the normal distribution with the mean of $f(x_{t-1}, u_t)$ and covariance of R_t . The relationship between the velocity and pose is a deterministic function presented as:

$$y_t = g(y_{t-1}, x_t) \quad (2)$$

which means that the current pose, y_t , is determined based on the previous pose, y_{t-1} , and the current velocity, x_t . The observation equation is a zero order Markov chain:

$$p(z_t | m_{c_t}, y_t, c_t) \sim \mathcal{N}(z_t; h(m_{c_t}, y_t, c_t), Q_t) \quad (3)$$

where z_t is the observation in time t . The symbol $\mathcal{N}$ stands for the normal distribution with the mean of

$h(m_{c_t}, y_t, c_t)$ and covariance of Q_t . The data association decision and landmark location are shown by c_t and m_{c_t} , respectively. In the *full SLAM* problem formulation, the final objective is to find the posterior of the entire path and the map [7]:

$$p(s_{1:t}, m | u_{1:t}, z_{1:t}, c_{1:t}) \quad (4)$$

where s is the robot pose state vector, $s = \{x, y\}$, $c_{1:t}$ stands for known data association and m is the map which is the set of landmarks:

$$m = \{m_1 \dots m_N\} \quad (5)$$

It is shown that this problem can be written in the factored form [17]:

$$p(s_{1:t}, m | u_{1:t}, z_{1:t}, c_{1:t}) \quad (6) \\ = p(s_{1:t} | u_{1:t}, z_{1:t}, c_{1:t}) \prod_{i=1}^N p(m_i | z_{1:t}, s_{1:t}, c_{1:t})$$

This factored form can be solved using one of the RBPF based approaches discussed in [5, 8]. The localization problem is solved using the particle filter. Each particle, Y_t , in this approach consists of a robot trajectory estimate and a set of estimators of the individual landmark locations. In order to use the Rao-Blackwellized Particle Filter (RBPF) approach, proper proposal and importance weight distributions have to be determined.

A. *Proposal Distribution* Based on the idea discussed in [8], the proposal distribution is chosen as:

$$s_t^{[k]} \sim p(s_t | x_{1:t-1}^{[k]}, y_{1:t-1}^{[k]}, u_{1:t}, z_{1:t}, c_{1:t}) \quad (7)$$

This distribution includes all information related to the path, $x_{1:t-1}^{[k]}$ and $y_{1:t-1}^{[k]}$, controls, $u_{1:t}$, observations, $z_{1:t}$, and data associations, $c_{1:t}$. Considering the fact that $s = \{x, y\}$, this distribution can be expanded as:

$$p(s_t | x_{1:t-1}^{[k]}, y_{1:t-1}^{[k]}, u_{1:t}, z_{1:t}, c_{1:t}) \quad (8) \\ = p(x_t | x_{1:t-1}^{[k]}, y_{1:t-1}^{[k]}, u_{1:t}, z_{1:t}, c_{1:t}) \\ \times p(y_t | x_t, x_{1:t-1}^{[k]}, y_{1:t-1}^{[k]}, u_{1:t}, z_{1:t}, c_{1:t})$$

Using Eq. (2) and the Markov rule:

$$p(y_t | x_t, x_{1:t-1}^{[k]}, y_{1:t-1}^{[k]}, u_{1:t}, z_{1:t}, c_{1:t}) \quad (9) \\ = p(y_t | x_t, y_{t-1}^{[k]}) = \delta_{y_t, y_t^d}$$

where δ is Kronecker delta and y_t^d is the deterministic result of the Eq. (3) when x_t and $y_{t-1}^{[k]}$ are given. Therefore, the proposal distribution of Eq. (8) is simplified as:

$$p(s_t | x_{1:t-1}^{[k]}, y_{1:t-1}^{[k]}, u_{1:t}, z_{1:t}, c_{1:t}) \quad (10) \\ = p(x_t | x_{1:t-1}^{[k]}, y_{1:t-1}^{[k]}, u_{1:t}, z_{1:t}, c_{1:t}) \delta_{y_t, y_t^d}$$

Thus, sampling must occur in two separate stages. The

velocity states, x_t , has to be drawn from the following probability distribution:

$$x_t^{[k]} \sim p(x_t | x_{1:t-1}^{[k]}, y_{1:t-1}^{[k]}, u_{1:t}, z_{1:t}, c_{1:t}) \quad (11)$$

whereas the states related to the robot pose, y_t , have to be calculated using Eq. (2). Using the Bayesian formulation, the probability distribution of Eq. (11) can be presented as:

$$\xrightarrow{\text{Bayes}} x_t^{[k]} \sim \eta^{[k]} p(z_t | x_t, x_{1:t-1}^{[k]}, y_{1:t-1}^{[k]}, u_{1:t}, z_{1:t-1}, c_{1:t}) \quad (12)$$

$$\times p(x_t | x_{1:t-1}^{[k]}, y_{1:t-1}^{[k]}, u_{1:t}, z_{1:t-1}, c_{1:t})$$

and using Markov chain:

$$x_t^{[k]} \sim \eta^{[k]} p(z_t | x_t, x_{1:t-1}^{[k]}, y_{1:t-1}^{[k]}, u_{1:t}, z_{1:t-1}, c_{1:t}) \quad (13)$$

$$\times p(x_t | x_{1:t-1}^{[k]}, u_t).$$

A general closed form solution of Eq. (13) is not available unless the linearizability assumption [8] is imposed on the system. Using these assumptions the closed form of the Eq. (14) can be obtained as the:

$$p(x_t | x_{1:t-1}^{[k]}, y_{1:t-1}^{[k]}, u_{1:t}, z_{1:t}, c_{1:t}) = \eta^{[k]} \exp\{-P_t^{[k]}\} \quad (14)$$

where

$$P_t^{[k]} = \frac{1}{2} [(z_t - \hat{z}_t^{[k]} - H_y(y_t - \hat{y}_t^{[k]}))^T Q_t^{[k]-1} (z_t - \hat{z}_t^{[k]} - H_y(y_t - \hat{y}_t^{[k]})) \quad (15)$$

$$- \hat{z}_t^{[k]} - H_y(y_t - \hat{y}_t^{[k]})]^T$$

$$+ (x_t - \hat{x}_t^{[k]})^T R^{-1} (x_t - \hat{x}_t^{[k]})]$$

where $\hat{z}_t$, $\hat{y}_t$ and $\hat{x}_t^{[k]}$ are prediction of the observation, velocity states and pose states, respectively. Also H_y is the Jacobian of the measurement model with respect to the pose states. In addition, $Q_t^{[k]}$ is defined as:

$$Q_t^{[k]} = Q_t + H_m \Sigma_{c_t, t-1}^{[k]} H_m^T \quad (16)$$

where H_m is the Jacobian of the measurement model with respect to the landmark location and $\Sigma_{c_t, t-1}^{[k]}$ is the covariance of the observed landmark location estimation. Considering the fact that the covariance of the Gaussian function is the inverse of the second derivative, and also invoking Eq. (16), the sampling distribution covariance is obtained from:

$$\Sigma_{x_t}^{[k]} = [(H_y Y_x)^T Q_t^{[k]-1} (H_y Y_x) + R_t^{-1}]^{-1} \quad (17)$$

where Y_x is the Jacobian of the Eq. (2) with respect to the robot velocity states. Also, the mean value which is the root of the first order derivative equals to:

$$\mu_{x_t}^{[k]} = \Sigma_{x_t}^{[k]} (H_y Y_x)^T Q_t^{[k]-1} (z_t - \hat{z}_t^{[k]}) + \hat{x}_t^{[k]} \quad (18)$$

In summary, to sample from the trajectory, the first velocity

states have to be drawn from the Gaussian distribution with the covariance and mean indicated in Eq. (17) and Eq. (18), respectively. Then, the pose states have to be calculated using Eq. (2).

B. Importance weight

As it is discussed in [8] in the particle filtering and also RBPF, resampling is the crucial stage of the filtering process. In order to resample the particle, a proper importance weight has to be assigned to each particle. As the other particle filter approach, the importance weight is defined by the following quotient:

$$\omega_t^{[k]} = \frac{\text{target distribution}}{\text{proposal distribution}} \quad (19)$$

The target distribution is the robot trajectory posterior, $p(s_t^{[k]} | u_{1:t}, z_{1:t}, c_{1:t})$. The proposal distribution is $p(s_t^{[k]} | s_{1:t-1}^{[k]}, u_{1:t}, z_{1:t}, c_{1:t})$ given the correct posterior from the previous stages, $p(s_{1:t-1}^{[k]} | u_{1:t-1}, z_{1:t-1}, c_{1:t-1})$. Therefore, Eq. (19) can be written as:

$$\omega_t^{[k]} = \eta p(z_t | s_{1:t-1}^{[k]}, u_{1:t}, z_{1:t}, c_{1:t}) \quad (20)$$

This equation can be transformed as:

$$\omega_t^{[k]} = \eta \int p(z_t | s_t, s_{1:t-1}^{[k]}, u_{1:t}, z_{1:t-1}, c_{1:t}) \quad (21)$$

$$\times p(s_t | s_{1:t-1}^{[k]}, u_{1:t}, z_{1:t-1}, c_{1:t}) ds_t$$

This equation can be simplified further using Markov chain to:

$$\omega_t^{[k]} \quad (22)$$

$$= \eta \int p(x_t | x_{t-1}, u_t) dx_t \int p(z_t | y_t, m_{c_t})$$

$$\times p(m_{c_t} | u_{1:t}, s_{1:t-1}^{[k]}, u_{1:t}, z_{1:t-1}, c_{1:t}) dm_{c_t}.$$

Using the system and measurement model, Eq. (1-3), and linearizing Eq. (2-3), the closed form of the Eq. (22), can be obtained as the Gaussian distribution:

$$\omega_t^{[k]} = |2\pi L_t^{[k]}|^{-\frac{1}{2}} \exp\left\{-\frac{1}{2} (z_t - \hat{z}_t^{[k]})^T L_t^{[k]-1} (z_t - \hat{z}_t^{[k]})\right\} \quad (23)$$

where

$$L_t^{[k]} = [(H_y Y_x)^T Q_t^{[k]} (H_y Y_x) + H_m^T \Sigma_{c_t, t-1}^{[k]} H_m \quad (24)$$

$$+ R_t]^{-1}.$$

The higher order RBPF for M particles can be summarized as it is shown in Table 1.

Table 1
Pseudo-code of higher order RBPF approach

- Do the following M times:
 - Retrieve the previous trajectory information $s_{t-1}^{[k]}$ from the k^{th} particle set of the previous iteration, $Y_{t-1}^{[k]}$ (*Retrieval*).
 - Sample a new velocity $x_t^{[k]}$ based on the proposal distribution (*Prediction*).
 - Update the pose based on the sampled velocity (*Prediction cont'd*).
 - Update the mean and covariance of the landmark distribution using the predicted pose and the observation of the landmark (*Measurement update*).
 - Calculate the importance weight $\omega_t^{[k]}$ for the new particle (*Importance weight*).
- Sample, with replacement, M particles, where each particle is sampled with a probability proportional to $\omega_t^{[k]}$ (*Resampling*)

III. MOTION AND MEASUREMENTS MODEL

A. Camera Motion Model

The camera motion is modeled using a procedure similar to the [18]. Using the constant velocity input assumption, the velocity update equation can be expressed as:

$$\begin{pmatrix} v_t^w \\ w_t^w \end{pmatrix} = \begin{pmatrix} v_{t-1}^w + V^w \\ w_{t-1}^w + \Omega^w \end{pmatrix} \quad (25)$$

where v_t^w and w_t^w are the velocity and rotational velocity of the camera, respectively. Also

$$u = \begin{pmatrix} V^w \\ \Omega^w \end{pmatrix} = \begin{pmatrix} a^w \Delta t \\ \alpha^w \Delta t \end{pmatrix} \quad (26)$$

where a^w , α^w and Δt are acceleration, angular acceleration and time interval, respectively. In the bearing-only situation acceleration and angular acceleration can be estimated by a Gaussian white noise. The camera pose can be defined by camera optical center, r^w , and the quaternion, q^w , defining the position and orientation of the camera in the global coordinates, respectively:

$$g = \begin{pmatrix} r^w \\ q^w \end{pmatrix}_t = \begin{pmatrix} r_{t-1}^w + v_t^w \Delta t \\ q_{t-1}^w \times q(w_t^w \Delta t) \end{pmatrix} \quad (27)$$

where $q(\Omega_k^w \Delta t)$ is the quaternion defined by the rotation vector $\Omega_k^w \Delta t$.

B. Measurement Model

In a 3D scene, the observation of every point i from a camera is defined by a vector consisting of six states:

$$m_i = (x_i, y_i, z_i, \theta_i, \phi_i, \rho_i)^T \quad (28)$$

This vector determines the point location as:

$$Y = \begin{pmatrix} x_i \\ y_i \\ z_i \end{pmatrix} + \frac{1}{\rho_i} n(\theta_i, \phi_i) \quad (29)$$

where $(x_i, y_i, z_i)^T$ is the location of the optical center of the camera at the first observation, ρ_i is the inverse of the point depth, d_i , along the ray, θ_i and ϕ_i are azimuth and elevation angles of ray directional vector at the global coordinate frame. Each observation can be interpreted as a constraint between the camera pose and the corresponding landmark.

Using the pin-hole model [19], the location of the landmark in the camera coordinate is given by:

$$h^c = R^{cw}(Y - r^w) \quad (30)$$

where $R^{cw}(q^w)$ is the rotation matrix from the global coordinate frame to the camera coordinate frame. The last equation can be expanded as:

$$h^c = R^{cw} \left(\rho_i \begin{pmatrix} x_i \\ y_i \\ z_i \end{pmatrix} - r^w \right) + n(\theta_i, \phi_i) \quad (31)$$

According to the pinhole model, the camera observes the projection of h^c into the image plane denoted by:

$$z = h(x, m) = \begin{pmatrix} u_0 - \frac{f}{d_x} v \\ v_0 - \frac{f}{d_y} v \end{pmatrix} \quad (32)$$

where u_0 and v_0 are the camera center in pixels, f is the focal length, and d_x and d_y are the pixel sizes. Also, v and v are defined as:

$$\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} \frac{h_x^c}{h_z^c} \\ \frac{h_y^c}{h_z^c} \end{pmatrix} \quad (33)$$

IV. FEATURE INITIALIZATION

The initialization process includes both initial estimation of the state values and assigning the related covariance matrix. Once a landmark is initialized, the estimation process is carried out using an EKF. The proposed initialization approach is similar to the initialization approach discussed by [20] but modified according to the RBPF approach used here. The initial state values for the landmark, $\hat{m}_i$, can be expressed as:

$$\hat{m}_i = (\hat{x}_i \ \hat{y}_i \ \hat{z}_i \ \hat{\theta}_i \ \hat{\phi}_i \ \hat{\rho}_i)^T \quad (34)$$

These values have to be determined using the location information, observation of new landmarks and the initial estimation of the depth inverse ρ_0 . The optic center of the camera is directly taken from the current estimation of the camera location. The azimuth and elevation angles of the projection ray are obtained from inverse of the camera pin-hole model. In the range $(0, \frac{1}{d_{min}})$, the inverse depth, $\hat{\rho}_0$, can be estimated as:

$$\hat{\rho}_0 = \frac{1}{2d_{min}} \quad (35)$$

The difference between initialization scheme of the higher order RBPF approach and initialization discussed in [20] lies in the initial covariance matrix used. In the higher order RBPF, the covariance matrix after this initialization is:

$$P = J \begin{pmatrix} D_t & 0 & 0 \\ 0 & Q_t & 0 \\ 0 & 0 & \sigma_\rho^2 \end{pmatrix} J^T \quad (36)$$

where Q_t is the noise covariance of camera, and the Jacobian is given by:

$$J = \left(\frac{\partial m}{\partial r_k^w}, \frac{\partial m}{\partial q_k^w}, \frac{\partial m}{\partial h}, \frac{\partial m}{\partial \rho} \right) \quad (37)$$

Also,

$$D_t = H_x R_t H_x^t \quad (38)$$

where H_x is the Jacobian of the pose states with respect to the velocity states and R_t is the control input covariance. The depth covariance is defined as:

$$\sigma_\rho = \frac{1}{4d_{min}} \quad (39)$$

V. SIMULATION RESULTS AND DISCUSSION

A simulation program has been developed to evaluate the performance of the proposed higher order RBPF vSLAM method. In order to appreciate the difference between the higher and lower RBPF performance, a simple straight path with trapezoid velocity profile has been examined. The velocity profile is composed of three sections including accelerating in the beginning, constant velocity in the middle, and decelerating in the end. In order to have a comparison between the performance of the lower and higher RBPF under erroneous control input assumptions, the control inputs (i.e., acceleration and velocity for higher and lower order, respectively) are overestimated by 50% in the simulation. The acceleration and maximum velocity are $0.4 (m/s^2)$ and $2 (m/s)$, respectively. Also, both filters consist of 30 particles. The robot path and the estimations of lower and higher order RBPF SLAM approaches are shown in Fig. 1. The black circles, solid line, dashed line and the dotted line are the landmarks, true robot path, the lower and higher order RBPF path estimations, respectively. The time history of the robot motion in x and y direction and their estimations are shown in Fig. 2-3, respectively. In order to verify the trend of the mapping uncertainty, the time history of the covariance norm of different landmarks is shown in Figure Fig.4. The path tracking of this approach has been compared to RBPF approach Fig. 2. As it is shown in the Fig. 2, the higher order RBPF outperforms the lower order RBPF approach while similar uncertainty has been assumed in the tuning of the filters. As it is shown in Fig. 3 and 4, the higher order RBPF shows better performance in both directions. The camera is posed in a way that the parallax angle produced by motion in the x-direction is smaller than the parallax angle produced by the motion in the y-direction. While the higher order shows better performance in both directions, the lower order RBPF shows relatively similar error in different directions. Therefore, it can be inferred that higher order RBPF performance strongly depends on the parallax angle and in the low parallax angle the lower order

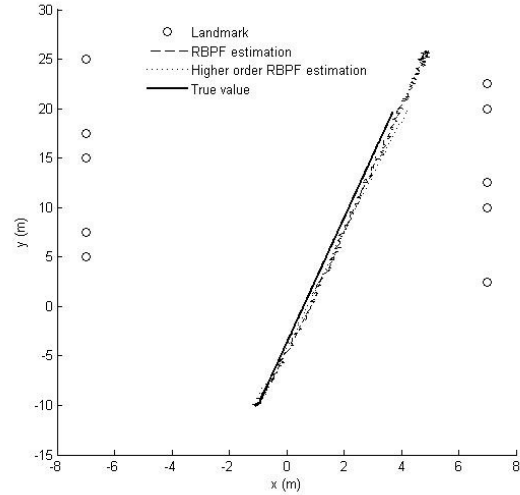


Fig. 1 Robot path, true value (solid line); RBPF estimation (dashed line); higher order RBPF estimation (dotted line).

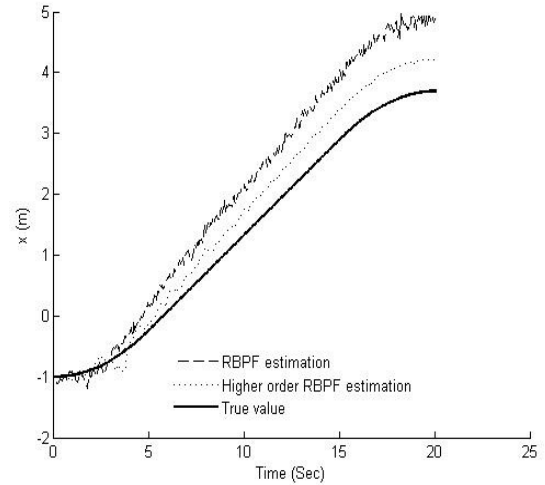


Fig. 2 The robot motion in x-direction, true value (solid line); RBPF estimation (dashed line); higher order RBPF estimation (dotted line).

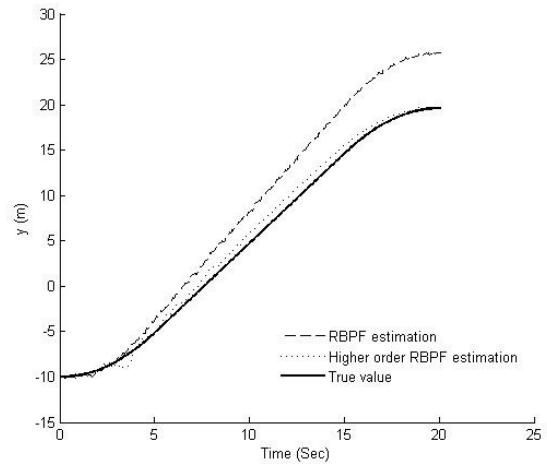


Fig. 3 The robot motion in y-direction, true value (solid line); RBPF estimation (dashed line); higher order RBPF estimation (dotted line).

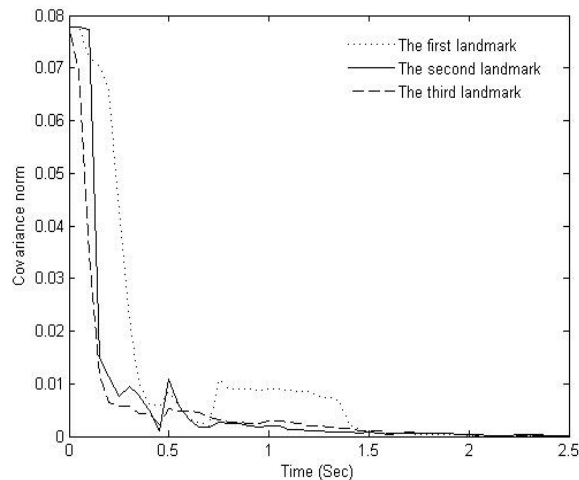


Fig. 4 The time evolution of the covariance matrix norms of the first three landmarks.

RBPF approach can outperform the higher order RBPF approach. As it is shown in Fig. 4, the covariance matrix norm, which is the expression of the uncertainty, generally reduces as the SLAM process progresses. While the covariance matrix norm reduction is generally correct, in some iteration the covariance matrix norm increases. This increase can be explained by the fact that in particle filtering, the particles with smaller importance weights are replaced with the particles with higher importance weights. Also, each of the particles in RBPF framework has its own map. Therefore, it is possible that a particle with a smaller uncertainty can be replaced by a particle with higher importance in an observation. Conclusion

VI. CONCLUSIONS

In this paper a new Rao-Blackwellized particle filter suitable for monocular vision (i.e., bearing-only) systems is proposed. In the proposed approach, unlike the traditional formulation where the velocity is considered a constant, filtering is performed under the constant acceleration assumption meaning that the velocity is a state variable estimated by the filter. Although the inclusion of higher order terms in a parameterized filter such as an EKF may be considered straightforward, the entire formulation must be reworked for such modification of RPBF. New sampling and importance weight distribution based on the constant acceleration assumption have been derived. Also to solve the bearing only problem, a modified initialization scheme has been proposed. Numerical simulations have been carried out to examine the performance of the proposed system. The higher order RBPF approach has been compared to a lower order RBPF approach in similar conditions. Simulation results verify not only the feasibility of the proposed higher order RBPF approach but also its superiority to the lower order RBPF approach for high parallax angle. It is also noted that the generalized higher order formulation is of great importance where there is a need to estimate the velocity as one of the states in the filtering process.

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